

1. According to the Empirical Rule, what percentage of data lies within:

- $\pm 1\sigma$
- $\pm 2\sigma$
- $\pm 3\sigma$

2. A dataset has a mean of 50 and a standard deviation of 10. According to the Empirical Rule, find the range where approximately 95% of the data lies.

3. A student scored 85 in an exam where the mean score is 70 and the standard deviation is 5. Calculate the Z-score of the student.

4. Given $Q1 = 25$ and $Q3 = 45$, calculate the IQR and explain what it represents about the dataset.

5. A dataset has a mean (μ) of 80 and a standard deviation (σ) of 5.

Using the Empirical Rule, determine the range where approximately 68% of the data lies.

6. The heights of students are normally distributed with a mean of 160 cm and a standard deviation of 8 cm. Find the range that contains approximately 95% of the students' heights.

7. A student scored 72 on a test where the mean score is 60 and the standard deviation is 6. Calculate the Z-score of the student.